Basics of Biotechnology

Basics of Biotechnology

- **Cell**
  - Basic unit of life
  - Has Nucleus, Cell membrane, cytoplasm and Cell organelles.
  - For growth and reproduction, important information needs to be transmitted.
  - Information is DeoxyRiboNucleic Acid (DNA)
  - Long DNA molecule needs to be condensed.
  - DNA to Chromatin to Chromosomes
- **DNA to RNA to Protein (Central Dogma)**
  - Transcription : DNA to mRNA
  - Translation : mRNA to Protein, tRNA and rRNA help in this process
  - Why RNA : Processing is easier
  - RNA vs DNA
    - Single Strand
    - Uracil in place of thymine
  - Why Protein : Enzymes for cellular function

Basics of Biotechnology

Basics of Biotechnology

- **Virus**
  - Borderline between living and non living
  - DNA / RNA and Protein but no cellular machinery
  - Can reproduce only inside host cell by integrating its genetic material
  - Some viruses start with RNA : Retro Virus
  - Reverse Transcription by Retro Virus
- **PYQs**
  - Can virus be cultured in artificial medium?
  - What genome does Retro virus have? DNA or RNA?
  - What is a transcriptome?
  - DNA Barcoding : can it be used to asses age?

Cell Biology and Human Reproduction

- **Cell Biology and Human Reproduction**
  - **Cell**
    - Fundamental unit of life as unicellular organisms can exist independently.
    - AV leeuwenhoek and Robert Brown
    - Cell theory
      - All living organisms are composed of cells.
      - Omnis Cellula e Cellula
    - Basic Structure of a cell : Nucleus, Cytoplasm, Plasma Membrane
    - Eukaryotic and Prokaryotic Cells based on nuclear membrane
  - **Cell Components**
    - Cellular Membrane
      - Fluid Mosaic Model
      - Phospholipid bi-layer
      - Protein channels, Integral and peripheral proteins
      - Selectively Permeable
      - Passive Transport
      - Active Transport
    - Cell wall
      - Prokaryotes, fungi, protista, plantae
      - Non living rigid structure
      - Shape, protection and cell to cell interaction
    - Cell organelles
      - Rough endoplasmic reticulum – Proteins and Transport
      - Smooth Endoplasmic Reticulum – Lipids and Transport
      - Lysosomes – Digestion
      - Golgi Apparatus – Packaging
      - Vacuole – Storage
      - Mitochondria – Powerhouse of the cell; Contains mitochondrial dna
      - Ribosomes – Protein formation
      - Chloroplast – Photosynthesis
      - Centrosome – Cell division in animals
    - Nucleus
      - Nucleolus : rRNA synthesis
      - Chromatin
      - Nucleoplasm
      - Nuclear pores

Prokaryotes vs Eukaryotes

	Eukaryotes	Prokaryotes
Definition	Clearly defined nucleus and cell organelles	no membrane bound organelles
Cell size	Bigger	Smaller
Organism Type	Multicellular and Unicelullar	Always Unicellular
Genetic Material	Inside membrane bound nucleus	Naked in the cytoplasm
Chromosomes	Multiple	One single loop of DNA and Plasmids
Ribosomes	Present, Larger	Present, Smaller
Cell wall	Simple cell wall in plants and fungi	Complex, all prokaryotes
Plasma membrane	Present	Present
Cytoplasm	Present	Present

Cell Biology and Human Reproduction

- **Cell Biology and Human Reproduction**
  - **Plant vs Animal Cells**
    - Chloroplasts
    - Cell wall
    - Vacuole
    - Centrioles
    - Centrosomes
  - **Human Reproduction**
    - Chromosomes are characteristic for a species
    - Chromosomes exist as a pair : diploid and haploid
    - Haploid cells : gametes : form offspring
    - Mitosis and Meiosis
      - Mitosis
        - 2 daughters cells
        - both diploid
      - Meiosis
        - 4 cells
        - all haploid
    - Gametes
      - Sperm (male)
        - Provides only haploid nucleus to the new zygote
      - Ovum (female)
        - Provides haploid nucleus, cytoplasm, mitochondria, and other organelles to the zygote
    - zygote and fertilisation
      - zygote has nuclei from both parents
      - organelles only from mother
      - formation of pronucleus
  - **3 parents' baby**
    - Mother's mitochondria is defective
    - Need to replace with healthy mitochondria
    - Method 1 : Pronuclear transfer, After fertilisation
    - Method 2 : Reconstructed egg technology : Before fertilisation
    - Zygote has DNA from 3 sources

Presented with xmind

Biotechnology

Principles of Biotechnology

- **Basics of Genetics**
  - DNA to RNA to Protein
    - Transcription
    - Translation
  - Gene vs Genome
    - Entire complement of DNA is Genome
    - Area coding for protein is Gene
    - Between 2 humans, 99.9% DNA is same. 0.1% Variation happens in non coding region.
    - Non coding region differences can be used for DNA fingerprinting.
    - During Transcription : Introns are removed.
- **Getting Better Genes**
  - Traditional Hybridisation
    - Uncertainty
    - Inclusion of Undesired genes
  - Genetic Engineering
    - Recombinant DNA technology
    - Joining 2 pieces of DNA generally not found together.
    - Production of Desired Proteins when integration happens
- **Basics of Biotechnology**
  - Using living organisms to produce products and processes useful to humans
  - 2 principles
    - Genetic Engineering
    - Bioprocess Engineering
  - Recombinant DNA
    - First time : Cohen and Boyer experiment in E. Coli
    - Using plasmid as vector
    - Use of restriction endonuclease
  - 3 steps of genetic modification
    - Identification of DNA with Desirable Genes
    - Introduction of Identified DNA into Host
    - Maintenance of introduced DNA in the Host and transfer to Progeny
  - Tools of RDT
    - Restriction enzymes : Recognise palindromic Sequnces
      - Formation of Sticky Ends
    - Vectors : ensure that Alien DNA can enter host

Biotechnology

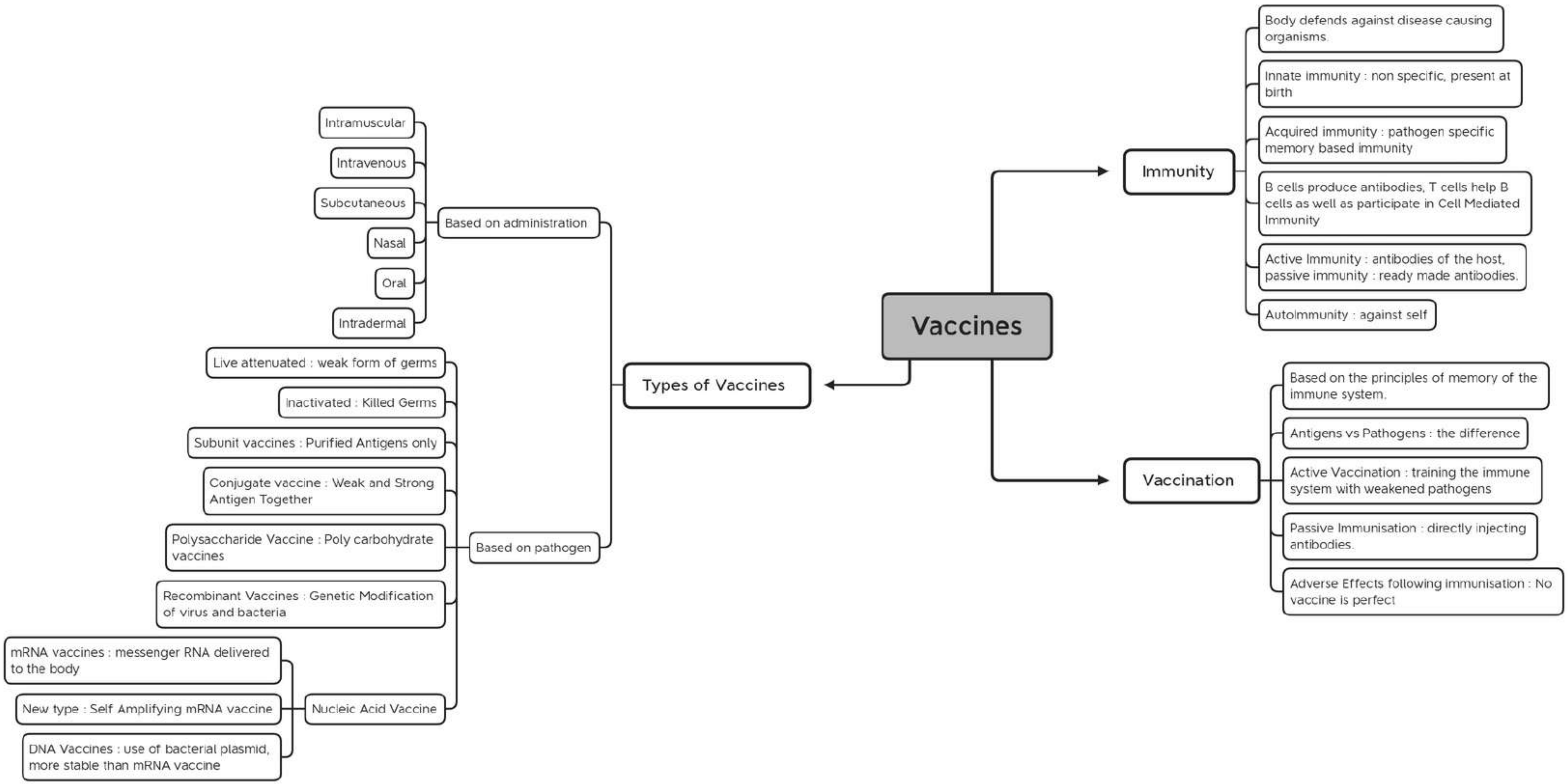
Principles of Biotechnology

- **Steps of RDT**
  - Isolation of DNA in Pure form
  - Cutting at specific places : Restriction Endonuclease
  - Polymerase Chain Reaction : Amplify gene of Interest
  - Ligation of 2 pieces of DNA
  - Insertion into host
  - Recombinant protein production
  - Downstream Processing
- **GM crops**
  - Pest Resistance : Use of BT cotton
    - Bacillus thuringensis provides the necessary genes
  - RNA Interference : Silencing of mRNA of Pathogen
    - Occurs in all eukaryotes
    - Using dsRNA to ensure pathogenic proteins do not get expressed
    - Used in Tobacco
    - Like a plant vaccination
  - Use of Mycorrhiza
    - Symbiosis between fungus and root
    - Provides benefits to plants
    - Absorptive area increased, drought tolerance, adverse conditions, pH, high temperature and heavy metals
  - Challenges :
    - Biosafety
    - Trait fees
    - Legal disputes
    - Terminator Genes
    - Genetic Divide
  - Regulation (more to be added soon)
    - GEAC
    - Ministry of Environment forests and climate change

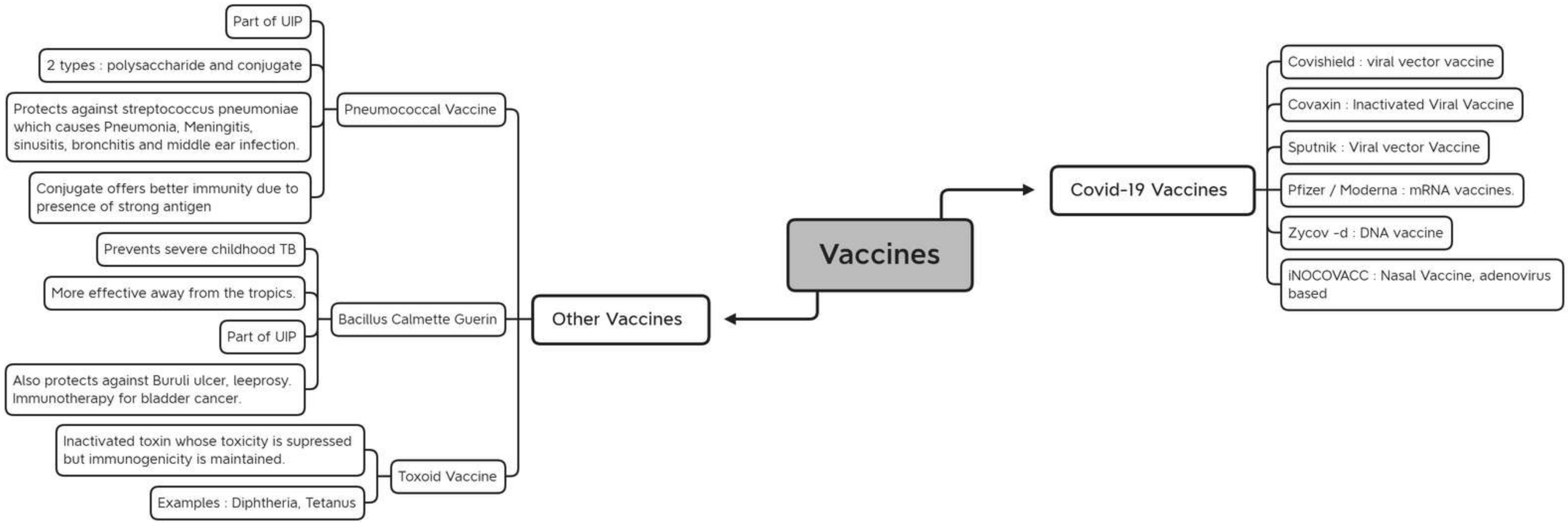
Biotechnology

- Principles of Biotechnology
  - Applications of Biotech Summary
    - Green Biotech
      - Yield
      - Bio pesticides : Bt cotton
      - HYV
      - Biofortification
      - Shelf Life
      - RNAi : plant vaccination
    - Blue biotechnology
      - Aquaculture
      - Transgenic fish
      - seaweeds
    - Red Biotechnology
      - Prevention
        - Vaccines
        - Transgenic Plants
        - Transgenic Animals
      - Diagnosis
        - Monoclonal Antibody
        - DNA probes
        - PCR
        - ELISA
      - Cure
        - Human Insulin
        - Gene therapy
        - Stem Cell Therapy
        - Recombinant proteins : Gamma interferon
    - Environmental Biotechnology
      - Oil zapper bacteria
      - Phytoremediation
      - DNA barcoding
    - Forensic Biotechnology
      - Identification of persons
      - Criminals / Paternity / Victims / Child Swapping
      - Use of DNA fingerprinting

Vaccines


Vaccines
Types of Vaccines
Based on administration
Intramuscular
Intravenous
Subcutaneous
Nasal
Oral
Intradermal
Based on pathogen
Live attenuated : weak form of germs
Inactivated : Killed Germs
Subunit vaccines : Purified Antigens only
Conjugate vaccine : Weak and Strong Antigen Together
Polysaccharide Vaccine : Poly carbohydrate vaccines
Recombinant Vaccines : Genetic Modification of virus and bacteria
Nucleic Acid Vaccine
mRNA vaccines : messenger RNA delivered to the body
New type : Self Amplifying mRNA vaccine
DNA Vaccines : use of bacterial plasmid, more stable than mRNA vaccine
Immunity
Body defends against disease causing organisms.
Innate immunity : non specific, present at birth
Acquired immunity : pathogen specific memory based immunity
B cells produce antibodies, T cells help B cells as well as participate in Cell Mediated Immunity
Active Immunity : antibodies of the host, passive immunity : ready made antibodies.
AutoImmunity : against self
Vaccination
Based on the principles of memory of the immune system.
Antigens vs Pathogens : the difference
Active Vaccination : training the immune system with weakened pathogens
Passive Immunisation : directly injecting antibodies.
Adverse Effects following immunisation : No vaccine is perfect

Vaccines


Vaccines
Covid-19 Vaccines
Covishield : viral vector vaccine
Covaxin : Inactivated Viral Vaccine
Sputnik : Viral vector Vaccine
Pfizer / Moderna : mRNA vaccines.
Zycov -d : DNA vaccine
iNOCOVACC : Nasal Vaccine, adenovirus based
Other Vaccines
Pneumococcal Vaccine
Part of UIP
2 types : polysaccharide and conjugate
Protects against streptococcus pneumoniae which causes Pneumonia, Meningitis, sinusitis, bronchitis and middle ear infection.
Conjugate offers better immunity due to presence of strong antigen
Bacillus Calmette Guerin
Prevents severe childhood TB
More effective away from the tropics.
Part of UIP
Also protects against Buruli ulcer, leeprosy. Immunotherapy for bladder cancer.
Toxoid Vaccine
Inactivated toxin whose toxicity is supressed but immunogenicity is maintained.
Examples : Diphtheria, Tetanus

Applications of Biotechnology

Applications of Biotechnology

- **GM Crops regulations**
  - Regulation
    - Global
      - UNCBD and 3 protocols
    - India
      - Environment protection act and 1989 rules
      - Competent Authorities
        - RDAC (advisory)
        - IBSC (approval, low risk)
        - RCGM (approval, high risk)
        - GEAC (final approval)
        - State Biotech Coordination Committee (monitoring)
        - District level Committe (monitoring, district level)
      - GEAC
        - Under MoEFCC
        - Gives approval for release, takes punitive action
        - Can revoke approval in certain cases
      - Exemption from rules
        - MoEFCC in certain cases
      - Event based approval
        - BT cotton
      - Role of Many ministries in Regulation
      - GM food imports : GEAC and FSSAI
- **GM crops**
  - Crops in india
    - Bt Cotton allowed
    - Bt Brinjal Banned
    - GM Mustard : environmental release allowed
    - GM Soymeal : Import Allowed
    - GM rubber : trials
  - Crops in the world
    - Flavr Savr Tomato (first commerical gm crop)
    - Golden Rice
  - Current Affairs
    - Agharkar research institute finds rht 14 and rht 18 in wheat : stubble burning issue can be solved
    - Exemption from 1989 rules to SDN 1 and SDN 2 edited crops
- **Recombinant DNA Technology**
  - Attaching 2 DNA that are generally not found together.
  - Genetic Engineering techniques
    - Genome Editing
    - Genetic Modification
    - Transgenesis
  - Applications
    - Transgenic Animals
    - Human Insulin Production

Applications of Biotechnology

- **Applications of Biotechnology**
  - **Gene Therapy**
    - Somatic and Germline Cells
    - Correction of Genetic Defects.
    - Somatic Gene Therapy and Germline Gene therapy
    - Use
      - Turn off a bad gene
      - Replace Bad gene with healthy gene
      - Add a new gene to perform function of missing gene
    - Types
      - in vivo
        - SMA type 1 (from video)
      - in vitro
        - CAR - T
        - ADA - SCID
    - CAR-T cell therapy to fight cancer
      - Developed by IIT bombay and Tata memorial center, now tested in Patients
      - Research funded under national biopharma mission by BIRAC
    - Cannot correct Chromosomal Defects, Short lived nature, Possible side effects, Virus Containment.
  - **Genome Editing Techniques**
    - Crispr Cas- 9
      - Cluster of Regularly interspaced palindromic repeat
      - Crisper Associated Protein - 9
      - Molecular scissors, Originally found in Bacteria
      - Guide RNA and Nuclease tailored for humans
      - Like ctrl + F in a document
      - Used for Precision guided sterile insects
      - Applications and Issues
    - Zinc Finger Nuclease
      - DNA scissors
      - Zinc Fingers for Recognition and FOKI nuclease for cutting
    - TALENs
      - Transcription activator like effector (TALE) for recognition
      - FOKI nuclease for cleaving
    - Site Directed Nuclease
      - SDN 1 : No foreign DNA
      - SDN 2 : Small Repair Template
        - (SDN 1 and SDN 2): Exemption from 1989 rules
      - SDN 3 : Foreign Gene enters
        - Not Exempted
    - Homing endonuclease

Applications of Biotechnology

- **Applications of Biotechnology**
  - **Animal Cloning**
    - Somatic Cell Nuclear Transfer Technique
    - Dolly the Sheep
    - Gene Cloning, Reproductive Cloning and Therapeutic Cloning
    - Recently, China has cloned Arctic Wolf
    - Therapeutic Cloning : healthy replica of human
      - Organ Transplant
      - Cancer
      - Aeging
      - Issues : ethical
  - **Stem Cell Therapy**
    - Cells having continuous ability to divide and replicate
    - Types
      - Totipotent
      - Pluripotent
      - Multipotent
      - Oligopotent
      - Unipotent
    - Induced Pluripotent stem cells : Yamanaka Genes
    - Used in cases of degenerative diseases, thalassemia, Leukemia, Osteoathritis, Osteoporosis, Blindness
    - Minimal Immune Rejection.
    - Stem Cell Banking, Future Prospects
    - Challenges : Synthetic Biology, Ethical Concerns

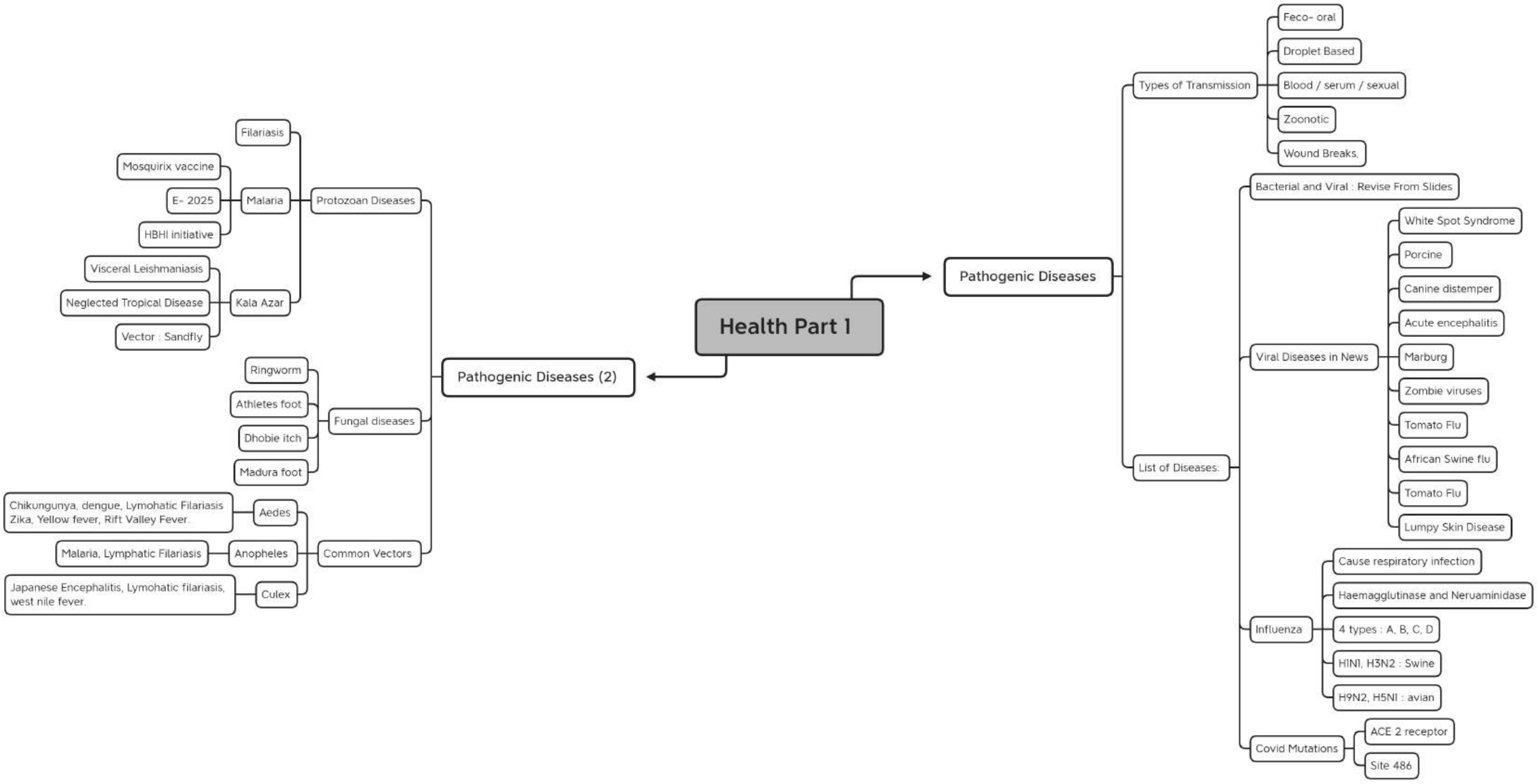
Health Part 1

- Health Part 1
  - Neglected Tropical Diseases
    - Communicable Diseases in tropical countries that dont get enough attention and funding
    - India has eliminated NTDs : Guinea worm, Trachoma, Yaws
    - Eradication of only one disease globally called smallpox
    - Control vs Elimination vs Eradication.
  - Traditional Medicine
    - New Global Center at Jamnagar
    - Traditional Medicine (indigenous) and Complementary Medicine (non indigenous)
    - Initiatives
      - National Ayush Mission
      - Champion sector for MVT
      - Ayush export promotion council
      - Ayur Swasthya Yojana
      - Ayush Information Cell
    - Ayurveda Ahaara regulations
      - FSSAI and MoAYUSH
      - Special logo
      - Ayurveda Drugs or proprietary medicines are not included
  - Genome Sequencing
    - Earlier effort : Human Genome project, 8% genome was left
    - Now : Telomere to Telomere project, complete sequencing done. Heterochromatin and Euchromatin
    - Advantages: disease diagnosis, Variation studies, Evolution, Gene therapy, viral diseases, pharmacogenetics
    - Indian Initiatives
      - Indi Gen project by CSIR
      - Genome India Project by DBT
  - Viruses
    - Borderline between living and non living.
    - Various types, RNA And DNA
    - Viroids : smaller infectious Particles and Free RNA
    - Plant viruses : ssRNA, Animals : dsRNA, dsDNA, Bacteriophage : dsDNA
    - Protein coat : capsid made of capsomeres.
    - Viral Vector vs DNA Vaccines

Health Part 1

- **Health Part 1**
  - **One Health Approach**
    - FAO, OIE, WHO and UNEP
    - Human Health, Animal Health and Environmental Health are interconnected as one health
    - Full spectrum of diseases : Prevention, Preparedness and Response.
    - Global
      - Pilanesberg resolution
      - Manhattan Principles
    - Indian Initiatives
      - G20 priority
      - Resource allocation : zoonotic diseases
      - National mission on biodiversity and well being (PM STIAC)
      - One health Project (DBT)
  - **AntiMicrobial Resistance**
    - Ability of microorganisms to stop working of antimicrobials.
    - Resistance levels continue to grow but antibiotic development is slow.
    - Causes
      - Poor hygiene
      - Overuse and non prescription use of antibiotics
      - Non completion of the anti microbial course
      - Role of environment (one health)
    - Indian Initiatives
      - National Programme on Containment
      - Red Line Campaign
      - Chennai Declaration
      - Schedule H1
      - Use of Phages
  - **Nutrient Deficiency Diseases**
    - Kwashiorkor : protein deficiency
    - Marasmus : protein and calories
    - Vitamins and their diseases
      - A : Retinol — Night blindness
      - D : Cholecalciferol — Rickets and osteomalacia
      - E : Tocopherol — Hemolytic Anaemia
      - K : Phylloquine — Bleeding
      - B1 : Thiamine — Beri Beri
      - B2 : Riboflavin — aRiboflavinosis
      - B3 : Niacin — Pellagra
      - B5 : Pantothenic Acid — Paraesthesia
      - B6 : Pyridoxine — Anaemia, Neuropathy
      - B7 : Biotin — Dermatitis
      - B9 : Folic acid — Neural Tube defects
      - B12 : Cyanocobalamine — Megaloblastic Anaemia
      - C : Ascorbic Acid — Scurvy

Health Part 1



Health Part 2

- **Health Part 2**
  - **Tuberculosis**
    - National Strategic Plan for TB elimination
    - Caused by mycobacterium tuberculosis
    - Vaccine only against childhood TB : BCG
    - Nikshay portal and Nikshay poshan yojana
    - PM TB mukt bharat by 2025
    - Patho detect kit
    - Moscow Declaration
    - BpAL : 6 month regimen
    - Related Term : DOTS
  - **Fats**
    - Saturated
      - Solid
      - No double bonds
      - Animal Origin, bad
    - Unsaturated
      - Liquid
      - Have many double bonds
      - Plant based, good.
    - Transfats
      - Solid @ Room temperature
      - Have one / few double bonds
      - Formed through partial hydrogenation, bad for you.
      - Increase bad cholesterol : LDL, reduce good cholesterol, HDL
      - Initiatives
        - Replace action framework
        - Part of triple billion WHO targets
        - India : TFA free logo
        - India : Heart attack Rewind

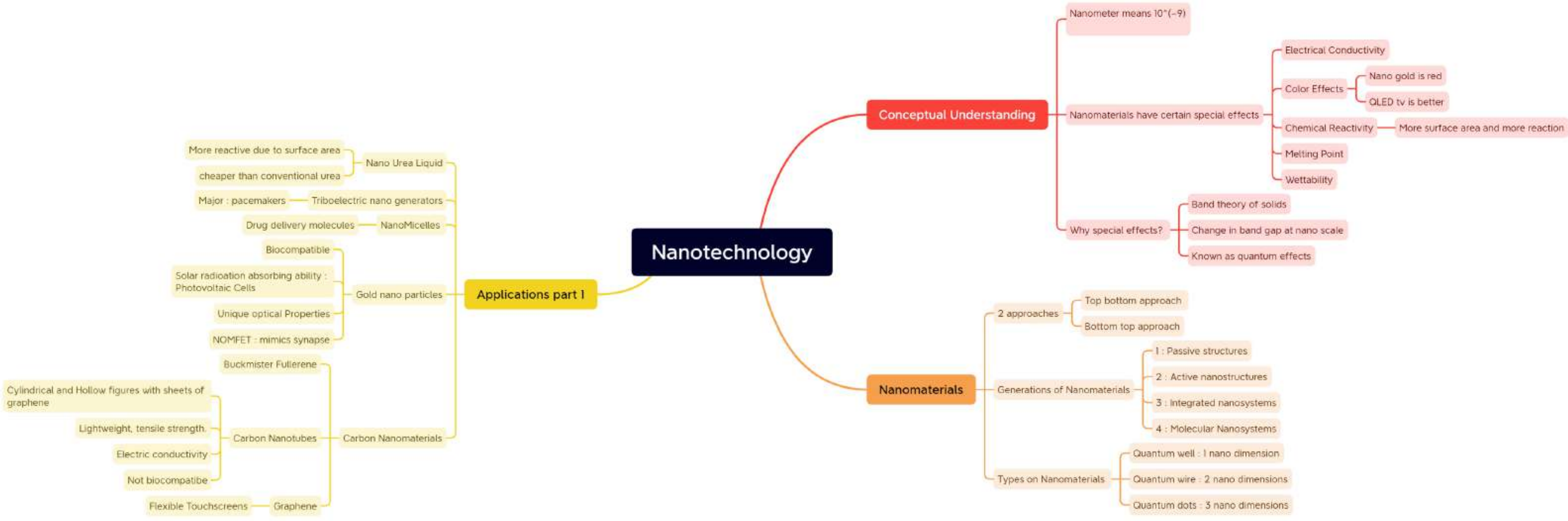
Health Part 2

- **Health Part 2**
  - **ORS**
    - Dr. Dilip Mahalanabis
    - Mixture of salts and carbohydrates dissolved in water.
    - SGLT co transporters
  - **Indian Biological data center**
    - Faridabad, Haryana
    - Supercomputer Brahm
    - FAIR data principles
    - Based on Biotech Pride Guidelines
  - **Genomic Surveillance**
    - Constant monitoring of the genome to look for any changes
    - Monitoring diseases
    - controlling pathogens
    - Developing vaccine
    - Targeted medicine
    - Indian Initiatives
      - Regional Genome Sequencing labs
      - Indian Sars cov-2 genomics consortium
      - Indian TB genomic surveillance consortium
  - **Some other diseases**
    - Dengue : aedes mosquito, vaccine developed
    - Sickle Cell Anaemia : Genetic Disease, Autosomal Recessive, National Mission this year.
    - Trachoma : elimination by malawi
    - Diabetes : type I (juvenile) and type 2 (adult), other types also
    - Shigella : causes shigellosis, diarrohea. No vaccine available.

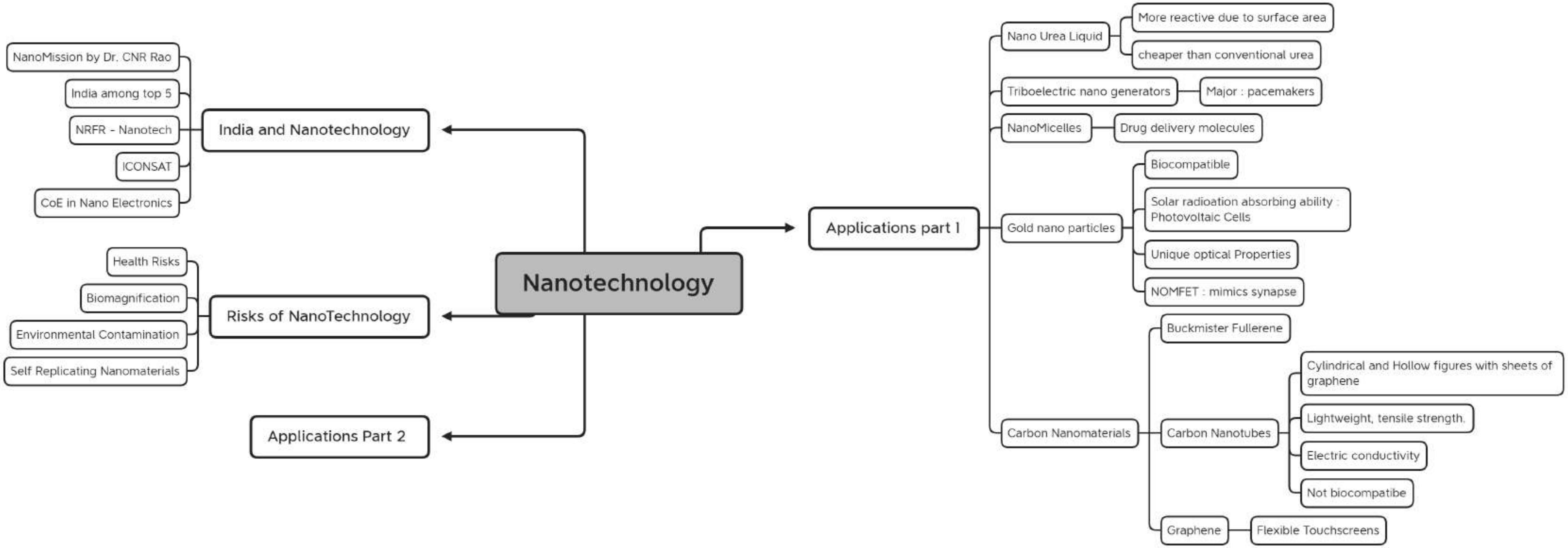
Health Part 2

- **Health Part 2**
  - **Foodborne Diseases**
    - Foodborne infections and Foodborne inToxication
    - Steps Taken by FSSAI for food safety
      - Eat Right Campaign
      - BHOG
      - Hygiene Rating scheme
      - Food safety compliance system
      - State food safety index
    - Antioxidants
  - **Food Fortification**
    - deliberately increasing the content of essential micronutrients, vitamins and minerals, ex : iodized salt
    - Regulation
      - Under food safety standard regulation
      - consumer affairs ministry : guidelines for fortified rice kernels : blend 1%
      - SOP for rice fortification
      - +F logo
    - Rice fortification
      - Mandatory : Iron, folic acid and Vit b12
      - Optional : zinc, vitA, thiamine, riboflavin, niacin, pyridoxine
    - Methods
      - Dusting
      - Coating and Extrusion
    - Benefits
      - Fight hidden hunger
      - Cost Effective
      - Deals with micronutrient deficiency diseases
      - No change in food habits
      - No change in characteristics of food
    - Forms
      - Mass
      - Targeted
      - Market driven
    - Steps taken by India
      - poshan abhiyaan and anaemia mukt bharat
      - FSSAI : food fortification resource center, Eat right india movement

Nanotechnology



Nanotechnology



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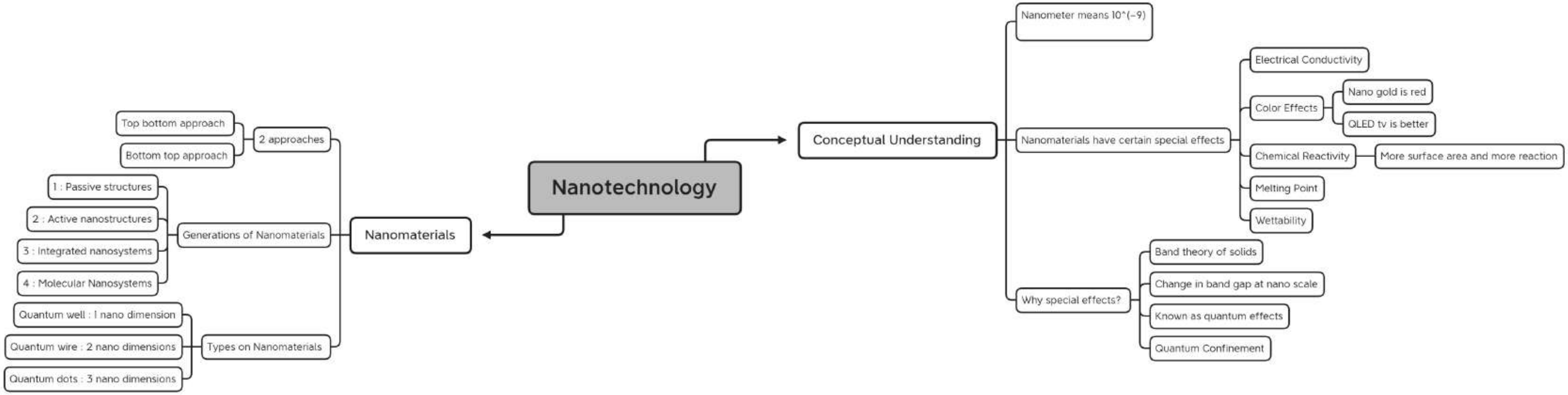
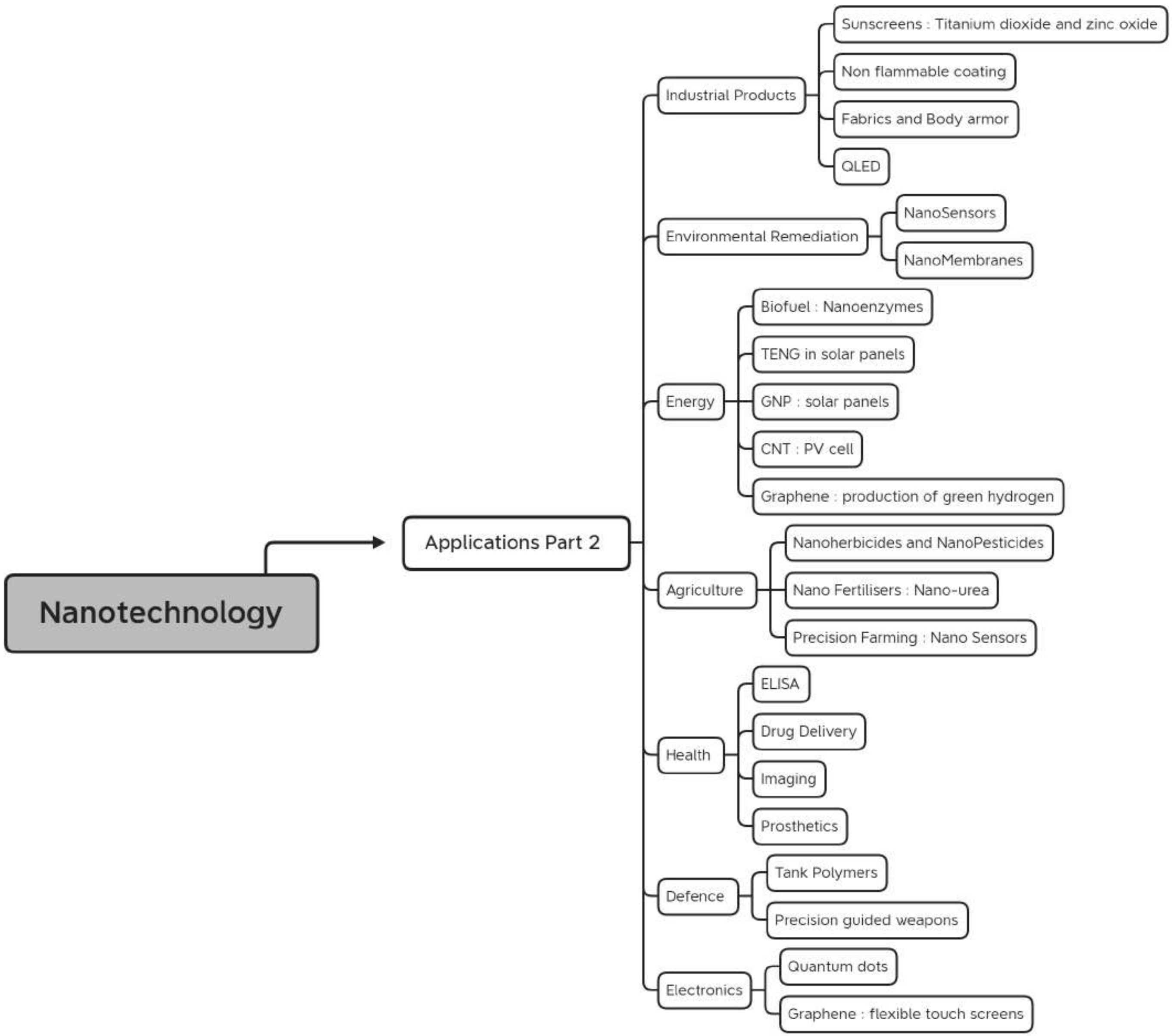


Nanotechnology



Space Technology

- **Space Technology**
  - **Space Tourism and miscellaneous**
    - Orbital vs Suborbital flight
    - Karman Line
    - Space sustainability
    - Retrograde orbit
  - **Relativity**
    - Basics of Stars
    - General Theory of Relativity
      - Gravity bends space time
      - Time is dilated by gravity
      - Light doesn't always travel in straight line
      - Special Theory of relativity : speed of light is fixed
      - Gravitational Waves
        - LIGO
        - Evolved LISA
        - Gravitational Lensing
    - Black Holes
      - Parts
        - Singularity
        - Accretion disc
        - Event Horizon
        - Schwarzschild radius
        - ErgoSphere
        - Relativistic Jet
      - Pictures of black holes
        - Event horizon telescopes
        - Meisser 87 and saggitarius A*
      - Tidal Disruption events
      - Binary Supermassive Black hole

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Space Technology

- **Space Technology**
  - **Space Debris**
    - both natural and manmade
    - Mostly in LEO
    - Techniques
      - Active removal
      - Move into Less populated orbits
      - Deorbiting at EOL
      - Orbital Lifetime reduction
    - Methods to Tackle
      - NETRA by ISRO
      - IIIT delhi project for SSA
      - UK tech demo sat I : re enter atmosphere and heat up
      - Elsa - D : retrieve space junk
      - Remove Debris Project of EU
        - NET capture
        - Harpoon Capture
        - De orbiting Process
        - Vision based navigation
    - Kessler Syndrome
  - **Indian Space Architecture**
    - ANTRIX
    - NSIL
    - InSPACE
    - Indian Space Association
    - Recent Private Sector initiatives
      - Use of POEM
      - Mission Prarambh

Space Technology

- **Space Technology**
  - **Space Exploration**
    - 3 components
      - Orbit
      - Lander
      - Rover
    - ISRO missions
      - Moon
        - Chandrayaan I
          - orbiter only
        - Chandrayaan 2
          - Orbiter
          - Vikram Lander (failed)
          - Pragyan Rover
          - GSLV MK 3 mission I
        - Chandrayaan 3
          - Only lander and rover
      - MARS
        - Mars orbiter mission I
          - First interplanetary mission
          - Tech and Scientific objectives
          - India became first asian country and 4th in the world
          - Payloads
            - MCC
            - LAP
            - TIIS
            - MENCA
            - MS
        - Other missions
          - Perseverance (USA)
            - 3 forks
          - Hope (UAE)
          - Tianwen I (china)
          - Mars moons exploration (japan)
      - Manned Mission
        - Gagan yaan
        - crew module and orbital module
        - Thermal protection system
    - Parker Space Probe
      - First Mission to touch the sun
      - Gravity procedures with Venus
      - 6 layers of sun
      - Solar Storms / Solar flares
        - Sun Spots
        - Solar Magnetic Cycle and Solar Dynamo Mechanism
        - Effects and Classification of Solar Flare
        - Subtopic 4
    - Artemis I
      - Space Launch System rocket
      - Orion Spacecraft
      - Ground systems at kennedy space center

Space Technology

- Space Technology
  - Satellites
    - Transponders
      - Various Frequency bands
        - L band — GPS
        - S band — Weather and surface ship radar
        - C band — Communication
        - X band — Military use
        - Ku band — Satellites and DTH
        - Ka band — Close range radars
      - Basics of Waves
        - EM spectrum
        - Wavelength and frequency
        - Concept of Rain fade
        - Relation of Antenna with wavelength
    - ISRO contributions
      - IRS
        - Resource SAT
        - Carto SAT
        - Bhuvan
        - Ocean SAT
        - NISAR
        - DM
      - GSAT
        - Communication
        - Weather
        - Tele education
        - Tele medicine
        - Disaster Management
      - GAGAN
        - 3 GSAT used
        - LPV gives similar performance to Category I ILS
        - AAI and ISRO : trials
        - Wide applications
      - Astrosat
        - Multi wavelength observatory
        - 650 km
        - Many other countries have this
        - Payloads
          - UVM
          - LAXPC
          - SXT
          - CZTI
          - SSM

Space Technology

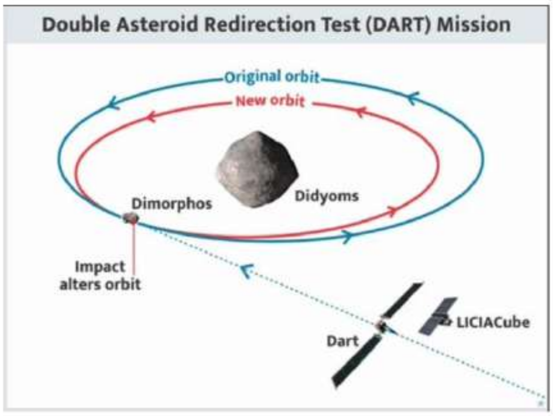
- **Space Technology**
  - Orbits
    - Definition : imaginary path taken by a spacecraft / celestial body around another celestial body
    - Classification
      - Shape
        - Elliptical
        - Circular
      - On basis of Revolution
        - Geo Synchronous
        - Sun Synchronous
    - Orbital velocity : specific to certain distance
    - LEO
      - 0-2000 km
      - Remote sensing
      - Astronomical Observation
      - Space Stations
        - ISS (global north, 5 countries)
        - Tiangong : only china
        - Applications : experimentation, observation, etc.
    - MEO
      - 2000 - 36000 km
      - Geostationary vs Geosynchronous
      - GPS systems
        - NAVSTAR
          - 24 satellites in semisynchronous orbit
          - Trilateralisation
        - GLONASS
        - Galelio
        - Beidou I and 2
      - Regional systems
        - Navic
        - QZSS
      - NAVIC
        - 4 geosynchronous, 3 geostationary
        - 10m resolution
        - 1500km from border
        - News : adding LI band to smartphones
      - GPS applications
        - Standard Positioning
        - Restricted Postioning
        - Timing services
    - High Earth Orbit
      - Lagrange points
      - LI, L2, L3 : unstable
        - JWST : L2
          - NASA, ESA, CSA
          - Infrared telescope
          - hot side and cold side
          - ISIM, OTE, Spacecraft element : 3 major payloads
        - Observations of JWST
          - Jupiter images
          - Pillars of Creation
          - Exo Moons
          - SMACS 0723
          - WASP 96b
          - Southern Ring Nebula
          - Cosmic Cliffs
          - Einstein ring
          - Stephan's Quintet
      - L4, L5 : stable
      - Geosynchronous Transfer orbit
        - used in mangalyaan

Space Technology

- **Space Technology**
  - **Launch Vehicles**
    - Expendable and Reusable Launch Vehicle
    - India's Launch vehicles (refer table in slides)
      - PSLV
        - Development of Poem
      - GSLV Mark 2
      - GSLV Mark 3
      - SSLV
    - Engines
      - Non air breathing
      - Air breathing
        - Ramjet : optimal speed mach 3 to mach 6
        - Scramjet : optimal speed mach 4 to mach 8
        - Dual Mode Ramjet
        - Solid Fuel Ducted Ramjet : Indo russian collaboration for missiles
      - Jet Engine : newton's 3rd law
    - Fuel
      - Solid Fuel
        - Maximum energy
        - Easy Storage
        - Heavy Weight
        - Non Uniform Burn
      - Liquid Fuel
        - Good efficiency
        - Less weight
        - Difficult storage : risk of leaks
      - Cryogenic
        - Liquid hydrogen and oxygen
        - Environment friendly

<table>
<tr><td>DART MISSION</td><td>

Objective→deflect an asteroid to change its motion in space and save earth from it's impact
DART's demo conducted on asteroid system Didymos and moonlet



</td></tr>
<tr><td>SOFIA Mission

Stratospheric Observatory for Infrared Astronomy</td><td>

SOFIA is an infrared telescope
Collaboration between NASA and German Space Agency
Objective→
understanding star birth and death
Formation of new solar systems
Discovery
Discovered helium hydride—>First molecule formed in universe almost 14 billion years ago

</td></tr>
<tr><td>ISRO</td><td></td></tr>
</table>

Hybrid propulsion system	• Unlike **solid-solid or liquid-liquid combinations, a hybrid motor uses solid fuel and a liquid oxidizer.** ○ The motor used **Hydroxyl-terminated polybutadiene (HTPB)** as fuel and **liquid oxygen (LOX)** as the oxidizer. • **Benefits**: more efficient, greener and safer to handle.
HS200 booster for Gaganyaan Programme	• It is the world's second largest **operational booster using solid propellants** • **Gaganyaan's objective** ○ indigenous capability to undertake **human spaceflight to Low Earth Orbit** (an orbit of 2,000km or less).

Inflatable Aerodynamic Decelerator (IAD)	• **ISRO has successfully tested IAD** technology to land **missions on Venus & Mars**.
	  ○ Used for **decelerating an object descending** through the atmosphere.
	• IAD is **made of Kevlar fabric** coated with polychloroprene.
	  ○ Kevlar has properties like **high tensile strength, toughness, thermal stability etc.**

Rohini Sounding Rockets	• **ISRO launched Rohini RH-200** sounding rocket.
	• Sounding rockets are
	    ○ **One or two stage solid propellant rockets**
	    ○ Used for **probing the upper atmospheric regions** and for **space research.**
	• **First sounding rocket** to be launched from India was **American Nike-Apache in 1963.**

RISAT-2 satellite	• **RISAT 2** is a **radar-imaging satellite**
	    • The RISAT satellites are equipped with a **synthetic aperture radar (SAR)**
	    • **Applications**
	        ○ **It can take pictures of the earth during day and night**, and also under cloudy conditions.
	        ○ It helps **round-the-clock border surveillance along with checking infiltration** and keeping an eye on terror or anti-national activities across the borders.

Aditya L1 mission	• Objective→ study of **Sun's corona, solar emissions, solar winds and flares,** and **Coronal Mass Ejections (CMEs)**
	• **Other solar missions:**
	  ○ European Space Agency's Solar Orbiter,
	  ○ NASA's Parker Solar Probe.

Space Tech Innovation Network (SpIN)	• **ISRO** has signed an MoU with Social Alpha **to launch SpIN.** • SpIN is India's **first dedicated platform space entrepreneurial ecosystem.**
SPACE PHENOMENON AND EXPERIMENTS	
Solar Eclipse	• A solar eclipse occurs when the Moon **gets between Earth and Sun**, and the moon **casts a shadow over Earth.** • Solar eclipses **happen only at the new moon phase.** • **Different types of solar eclipses are:** ○ **Total solar eclipse:** When moon

	passes between Sun and Earth, **completely blocking the face of Sun**. ○ **Partial solar eclipse:** When sun, moon and Earth are not exactly lined up. Only a part of the Sun will appear to be covered, **giving it a crescent shape.** ○ **Annular solar eclipse:** When the moon is farthest from Earth. It does not block the entire view of the sun. This **looks like a ring around the moon.** ○ **Hybrid solar eclipse:** Because Earth's surface is curved, sometimes an eclipse can shift between annular and total as the Moon's shadow moves across the globe.
Blood Moon	• **Blood Moon**, commonly known as **total lunar eclipse**, occurs when **moon passes through darkest part of Earth's shadow**, known as the **umbra.** ○ It is called **blood moon because of reddish hue.** • **Other Types of Lunar Eclipse:** ○ **Partial Lunar Eclipse:** When only a part of moon enters earth's shadow.

	◦ **Penumbral Lunar Eclipse:** When the moon enters the Earth's penumbra.
Geomagnetic Storm	• GMS is a **disturbance in the earth's magnetosphere**, which is the area around the planet controlled by its magnetic field. ◦ **When Coronal Mass Ejections** (CME) **collide with the Earth**, it causes GMS. ◦ CME is a **large expulsion of plasma and magnetic field from the sun's corona**, or upper atmosphere. • **Impact of geomagnetic storms—>PYQ** ◦ Disrupt high-frequency radio broadcasts and global positioning system (GPS) devices, damage satellite electronics, can affect power supply on earth.
Coronal Holes	• **Coronal holes** are **regions on the sun's surface from where fast solar wind gushes out into space.** • These coronal holes can **cause a solar storm on Earth** as they **release a complex stream of solar winds.**
Sunspots	• **Sunspots are areas that appear dark** on the surface of the Sun.

	◦ They appear dark **because they are cooler than other parts of the Sun's surface.** ◦ They **form at areas where magnetic fields are particularly strong** thus keeping some of the heat within Sun from reaching the surface. • **Magnetic field lines near sunspots can cause** a sudden explosion of energy called a **solar flare**.
Fast Radio Bursts (FRBs)	• FRBs are **bright bursts of radio waves whose durations lie in the millisecond-scale,** because of which it is difficult to detect them and determine their position in the sky. ◦ **Produced by astronomical objects with changing magnetic fields**
Gamma Ray Bursts (GRBs)	• GRBs are **flashes of high-energy radiation arising from energetic cosmic explosions.** ◦ GRBs are the **most powerful explosions the universe** has seen **since the Big Bang.** • Astronomers recorded **binary merger emitting long GRB twinned with a kilonova emissions.**

	○ Kilonova occurs when two compact objects, like binary neutron stars or a neutron star and a black hole, collide.
Betelgeuse	• It is the **bright red supergiant** on the shoulder of Orion. ○ In 2019 Betelgeuse **likely underwent an enormous surface mass ejection (SME)** ○ It ejected **400 billion times more mass than a typical event on other stars.** ○ SME happens when a star **expels large amounts of plasma and magnetic flux** into the surrounding space. • **A red giant is a dying star** in the final stages of stellar evolution.
White Dwarf	• White dwarfs are **stars that have burned up all of** hydrogen they once used as nuclear fuel.

Plants in Moon soil	• Scientists have for the first time successfully **grown plants in the Moon's soil**. • This lunar soil, also called **regolith**, was brought to Earth from the Moon by the **Apollo-era astronauts.**
SPACE OBJECTS	
New asteroid 2022 AP7 discovered	• Astronomers discovered a new **Potentially Hazardous Asteroid** named **2022 AP7.** • It **was 1.5 kilometer wide**. Any asteroid over 1km in size is considered a **planet killer.** • It will cross Earth's orbit
Exoplanets	Detection of **barium** in the upper atmosphere of two giant exoplanet (WASP- 76b and WASP-121b) **for the first time.** WASP-76b and WASP-121b are **Ultra-hot Jupiters, a class of hot gaseous planets that matches the size of Jupiter**.
Bernardinelli-Bernstein comet	• **largest icy comet nucleus ever seen by astronomers**. • BBC ○ Originated in **the Oort cloud,** a distant region of the solar system that is **predicted to be the source of most comets**.
MISC	

Dark SKY Reserve	• **India's first dark sky reserve at Hanle** in Ladakh as a part of **Changthang Wildlife Sanctuary.**
	  ○ World's **highest-located sites** for optical, infra-red, and gamma-ray telescopes.
	• **Dark Sky Reserve**
	  ○ Place with policies to ensure **minimal artificial light interference.**
	  ○ **International Dark Sky Association**, a U.S.-based non-profit, **designates places** as International Dark Sky Places

Sampurnanand Optical Telescope (SOT)	• Located at Nainital
	• SOT has been **used for optical observations of comets, occultation by planets and asteroids, star forming regions** and star clusters, active galactic nuclei, etc.
	• **Important discovery made:**
	  ○ Discovery of rings of Neptune;
	  ○ Contributed to detection of rings around Uranus and additional rings around Saturn;
	  ○ First detection of optical afterglows of Gamma-ray-bursts (GRBs);
	  ○ Microlensing event;
	  ○ Discovery of quakes in various stars (under Nainital-Cape Survey program) etc.

Shaped Antenna measurement of	• SARAS 3 is a **precision radio telescope** to detect extremely **faint radio wave signals from Cosmic Dawn.**

the background Radio Spectrum (SARAS 3) Telescope	• It is indigenously **designed and built at Raman Research Institute** • Located at **Dandiganahalli Lake** and Sharavati backwaters, **in Karnataka.** • SARAS 3 has **helped determine properties of the earliest radio luminous galaxies** formed 200 million years after the Big Bang, a period known as the Cosmic Dawn.

Liquid-Mirror Telescope (LMT)	• **India's first and Asia's largest liquid-mirror telescope**
	• commissioned **at the Devasthal Observatory** of Aryabhatta Research Institute of Observational Sciences (ARIES) **in Nainital (Uttarakhand).**
	• LMT will **observe asteroids, supernovae, space debris** and all other celestial objects.
	  ○ **Rotating mirror made up of a thin film of liquid mercury to collect and focus light.**
	  ○ Primary mirror is liquid
	  ○ It **cannot be turned and pointed in any direction**
	  ○ **It** watches the sky as the Earth rotates.
	• Built by astronomers from **India, Belgium and Canada**

Space Bricks	• **Bricks of complex shapes from Martian soil** with the help of bacteria and urea.
	  ○ Made by ISRO and IISc
	• Made by mixing Martian soil with
	  ○ **guar gum**,
	  ○ **a bacterium** called Sporosarcina pasteurii,

	○ **urea and** ○ **nickel chloride (NiCl2).**
Tiangong	• Space station built by China • 2nd such body in **low-Earth orbit** after the **NASA- led International Space Station (ISS).** • Tiangong will be **much smaller and lighter than ISS.**

Missile Facts

Defence Related Facts : Indian Missiles

Missille	Type and Details	From - to	Range (km)	Remarks
Prithvi I	SRBM	S2S	150	India's first indigenous missile, Nuclear capable, not in service
Prithvi II	SRBM	S2S	350	In service
Prithvi III	SRBM	S2S	600	In service
Dhanush	Naval variant of Prithvi II	Sea to Surface	350	In service
Agni I	MRBM, Single Stage	S2S	1250	In service
Agni II	IRBM, Two Stage	S2S	2000	In service
Agni III	IRBM, Two Stage	S2S	3000	In service
Agni IV	IRBM, Two Stage	S2S	4000	In service
Agni V	ICMB, 3 stage	S2S	5000	In service
Agni VI	ICBM, 4 stage	S2S	5500	In development
Agni P (prime)	maneuverable reentry vehicle, can be used as ASBM	S2S	Details pending	Tested in 2021
Trishul	Navy, short range	SAM	9km	Not in Service
Akash	Supersonic, Medium Range	SAM	30km	Recently, export authorised to friendly nations

Missile Facts

Missille	Type and Details	From - to	Range (km)	Remarks
Astra	Active Radar Homing, beyond visual range	A2A	80	First A2A by DRDO, Mach 4.5 speed, Solid fuel, has radar transceiver
Shaurya	Canister based, Medium Range, Hypersonic, tactical	S2S	700-1900	In service, land based parallel of submarine launched k-15
Prahaar	Tactical Ballistic Missile, QR, Omnidirectional Warhead	S2S	150	In service, meant to replace Prithvi 1
K4	SLBM, intermediate range, in development	Underwater to Surface	3500	Named after Dr. Kalam, Nuclear triad and second strike capability
Sagarika (k-15)	SLBM	Under water to surface	700	SLBM version of Shaurya, to be integrated with INS arihant
Barak 1	Short range SAM, ship defense	Ship to air, surface	12km	Built with Israel, to be integrated with Shivalik Class frigates
Barak 8	SRSAM, ship defense	Ship to air, ship to suface	90km	Israel, to be integrated with Kolkata class destroyers.
Nirbhay	Stealth, subsonic, cruise, nuclear capable	Land, naval, air	1000km	In service, no MTCR issues

Defence Technology

- Defence Technology
  - Missile Technology
    - Guided airborne ranged weapon capable of self propelled flight
    - Range : Land > Air > Sea Launched
    - Range
      - Short ≤ 500
      - Medium ≤ 1500
      - Intermediate ≤ 5000
      - Long Range (ICBM) ≥ 5000
    - Launch site and Target
      - Surface to Surface
      - Surface to Air
      - Air to Air
    - Speed
      - Subsonic
      - Trans- sonic
      - Supersonic
      - Hypersonic
    - Trajectory
      - Ballistic
        - Predetermined Target
        - Most of the flight is unpowered and unguided
        - exoatmospheric and endoatmospheric
        - Radar can easily detect it
      - Cruise
        - Guided missile
        - Full flight is powered
        - within the earth's atmosphere
        - Difficult to intercept by radar

Defence Technology

- Defence Technology
  - Missiles Continued
    - Ballistic Missiles
      - Tactical — 150 - 300
      - Theatre — 300 - 3500
      - SRBM — 300 - 1000
      - MRBM — 1000 - 3500
      - IRBM — 3500 - 5500
      - ICBM — ≥5500
    - ICBM
      - 5500 km
      - MIRV
    - Hypersonic Missiles
      - Mach 5, 1.6km/s
      - Midway manuevre
      - Hypersonic Cruise Missile
      - Hypersonic Glide Vehicle
      - FOBS
        - could fly over the south pole
        - Put in stable orbit
    - Missile Defence Systems
      - Intended to shield against incoming missiles
      - France, UK and Italy — PAAMS
      - France and Italy — Aster Missiles
      - Israel — Arrow, Iron Dome, Iron Beam
      - USA — THAAD
      - China — KT, HQ
      - Russia — S400, S500
    - India's Ballistic Missile Defence System
      - Pradhyuman Air Defence
        - exo-atmospheric
        - 80km altitude
        - 2 stage : solid fuelled and liquid fuelled
        - Guidance with Active Radar Homing
      - Advance Air Defence
        - 40 km altitude
        - endo-atmospheric
        - single stage : solid fuelled
      - S-400
        - LRSAM
        - aircraft, cruise, ballistic, restructured against ground targets, AWACS planes
        - 400km, 250km, 120km, 40km
        - Engage 80 targets at one time and response time of 9-10 seconds
    - Brahmos
      - India and Russia
      - Land, air, submarine and ships
      - 2 stage : first solid and liquid
      - 2.8 mach : supersonic
      - Fire and Forget
    - Agni Prime
      - Surface to Surface ballistic missile
      - 2 stage canisterised solid propellant
    - Supersonic Missile Assisted Torpedo
    - Other Missiles : Refer to Table

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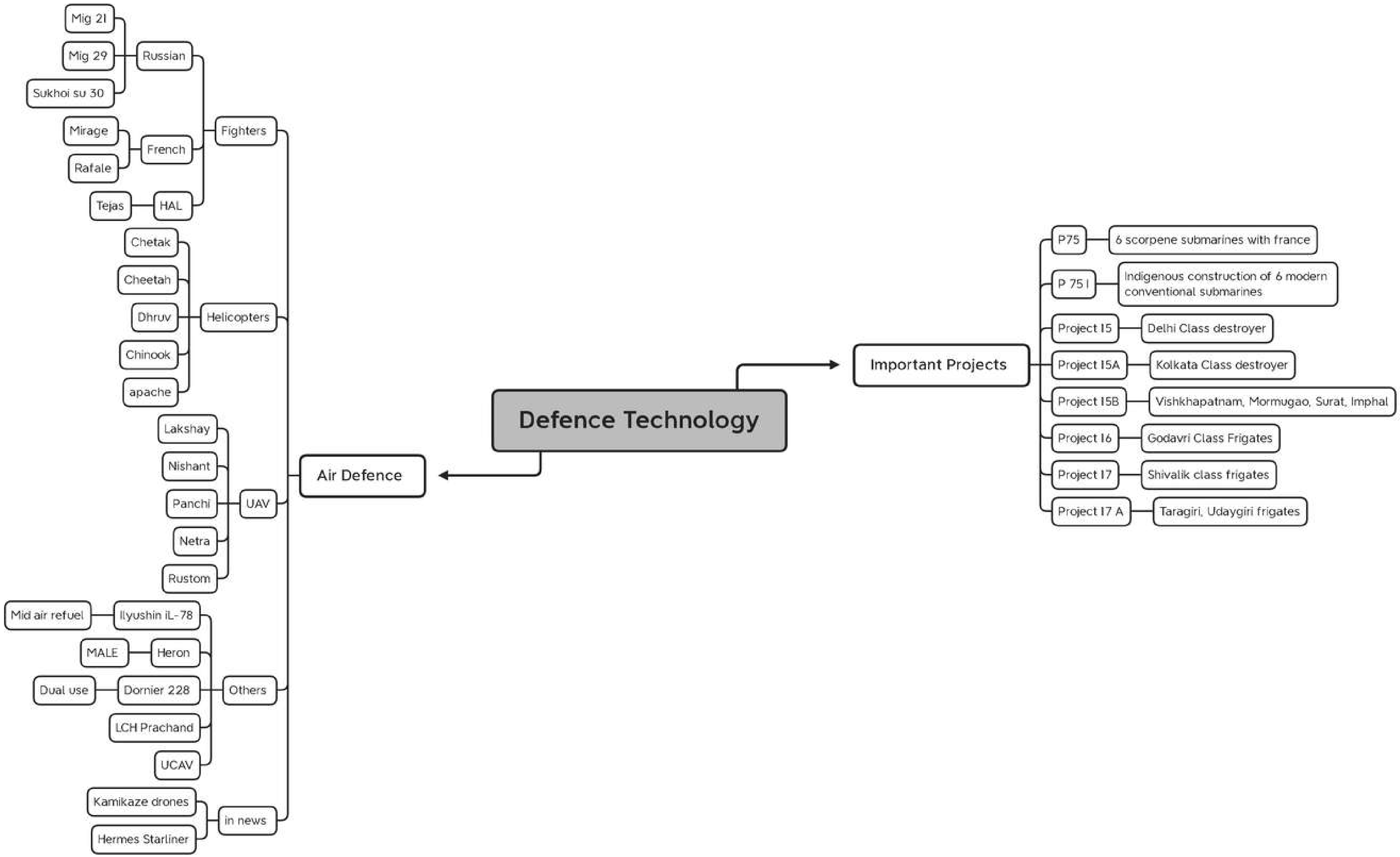
Defence Technology



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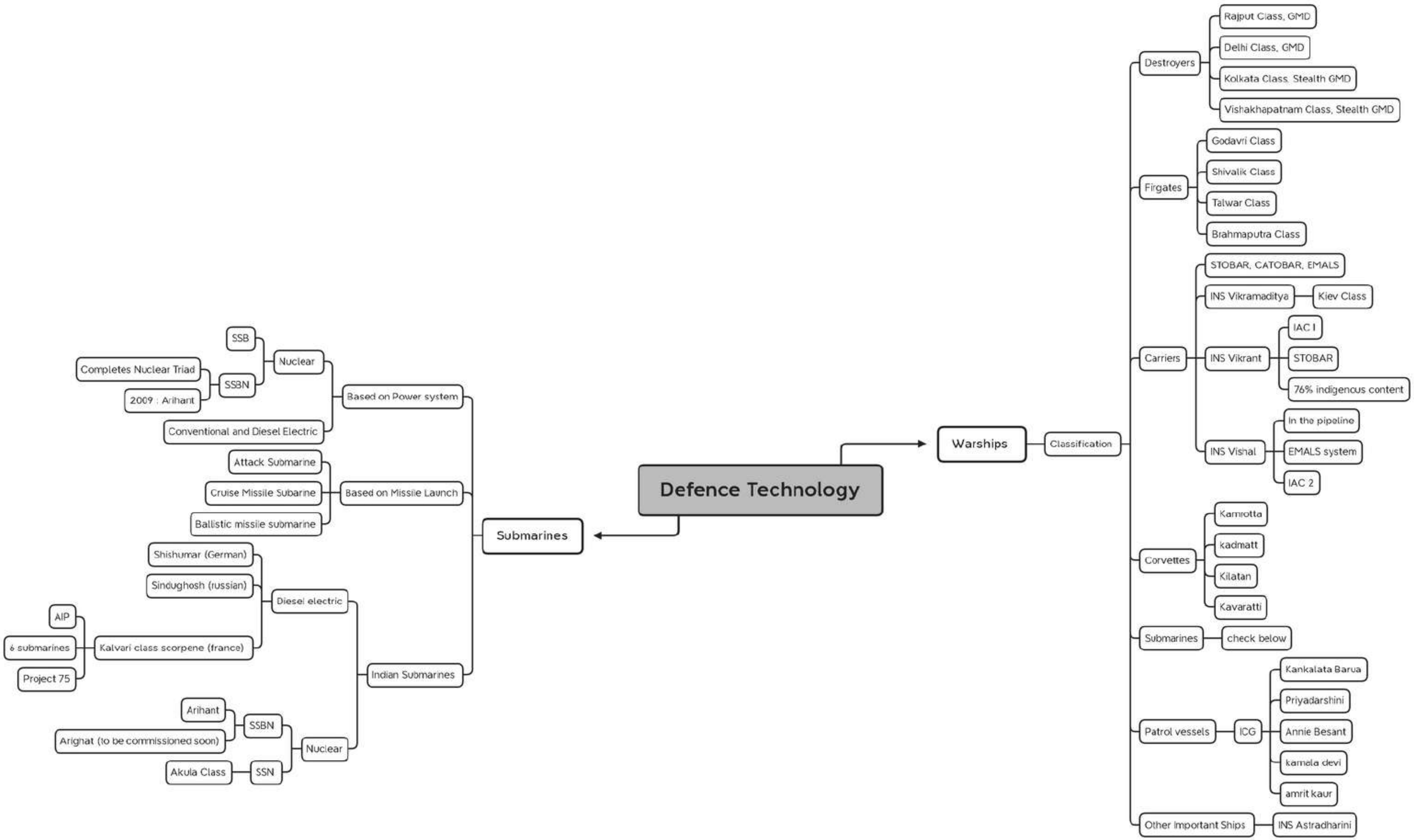

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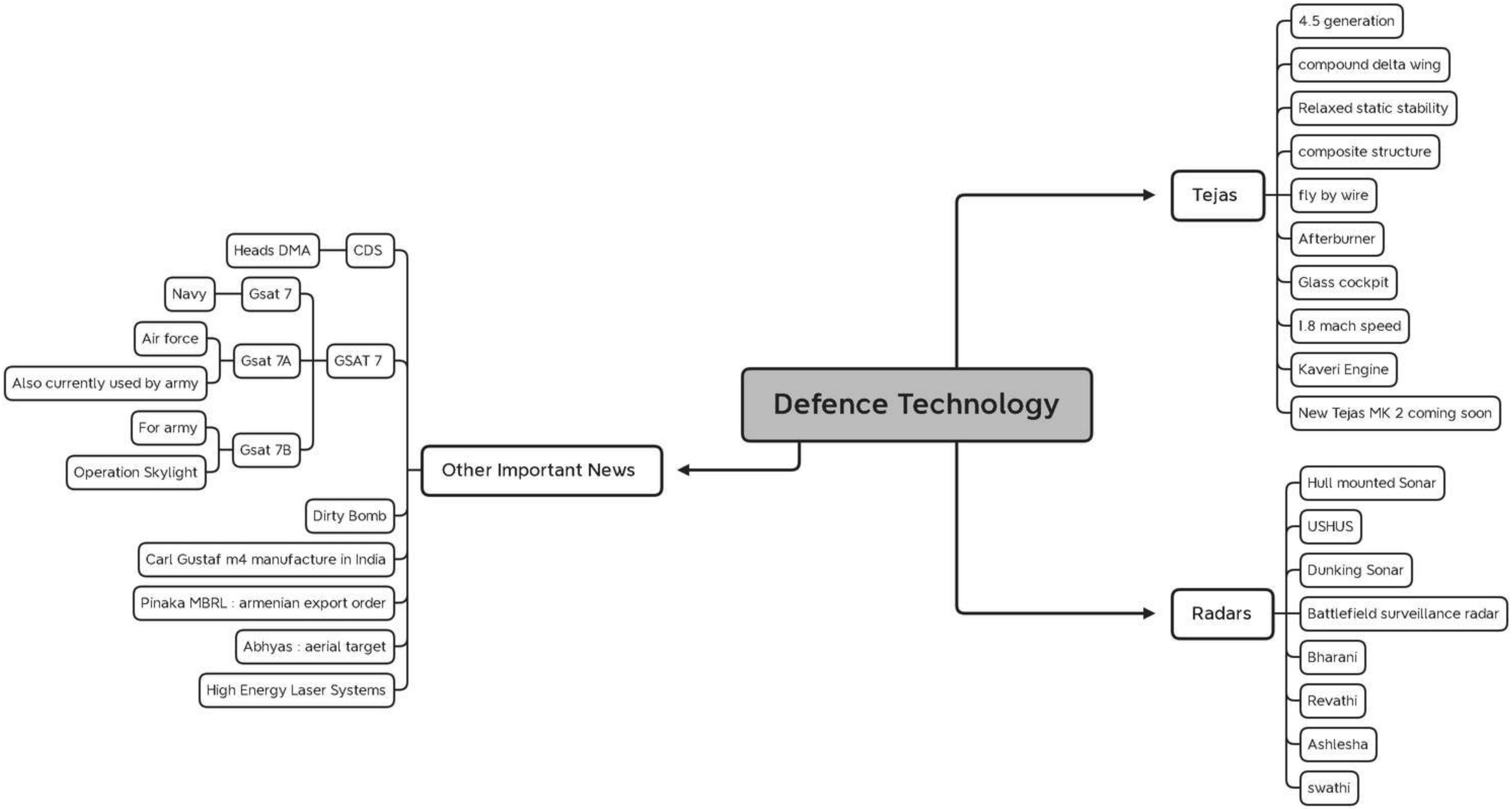

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Emergent Technology Part 1

Emerging Technology — Artificial Intelligence

- Simulation of human learning processes by machines : Learning, reasoning and self correction.
- Learning : create algorithms, Reasoning : choose the right algorithms, self correction : fine tune algorithms for the most accurate results.
- AI : subset : Machine Learning : Subset : neural networks : Subset : deep learning.
- Technologies in AI : Computer vision, audio processing, Natural Language processing, Knowledge representation, Machine learning and expert systems.
- Generative AI : can create new and original content indistinguishable from human made content. It is trained on large datasets.
- Generative adversarial networks, VAEs and Transformer Based models are 3 types of generative AI frameworks.
- Transformer based model : uses a deep neural network, an encoder and a decoder. Uses Natural Language processing to understand human language.
- Basic Idea is pattern analysis by bringing all sentences to similar frameworks.
- Steps of NLP : Tokenisation, Removing Stop Words, Stemming, Lemmetisation, Speech Tagging, Named Entity Recognition.

Emergent Technology Part 1

- **Emerging Technology**
  - LiFi
    - Light fidelity, use of visual light for communication with light emitting diodes.
    - Higher bandwidth than Wifi due to higher frequency.
    - Not a strictly line of sight technology, data rate is dependent on signal quality of device.
    - No interference due to higher bandwidth, coverage distance is less than wifi and data density is more.
  - 5G
    - 5th generation of cellular networks, solves the legacy issues of speed, latency and utility.
    - Speed is more than earlier generations while latency is less. Higher throughput (data transfer per unit second)
    - Can connect everyone and everything.
    - 3 bands : low, mid and high band and their respective uses.
    - Related terms
      - MIMO : multiple input multiple output
      - Small Cell Stations
      - Mobile Edge computing.
      - Beamforming
      - Network Slicing
      - Fiberisation.
      - 5G airwave interference.
    - Open Radio Access Network
      - CDOT has signed an agreement with private entities.
      - Will allow vendors to mix and match devices to be used for 5G.

Emergent Technology Part 1

Emerging Technology

- **Web 3.0**
  - Web 1.0 : Static Web
  - Web 2.0 : interactive web with targeted advertising.
  - Web 3.0 : blockchain based decentralised network where no one own the data, use of AI and machine learning.
  - NFTs, cryptocurrency, blockchain distributed across various networks.
  - Decentralised autonomous organisations, decentralised finance, decentralised apps.
  - Trustless and permissionless model with token based economics.
  - Web 5.0 : mixture of web 3.0 and web 2.0
- **NFTs**
  - Digital identifier and certificate of ownership of a virtual asset
  - Based on blockchain technology
  - No subdivision and no mutual substitution is possible.
  - Gives an opportunity to digital creators to monetise their assets, verify ownership records.
- **Internet of Things**
  - Machine to machine communication without any human intervention among devices to make our lives easier.
  - Phones, tablets and PCs are not part of IOT.
  - Web enabled smart devices that share sensor data.
  - Various applications and makes our lives simple.
  - Issues : need of high bandwidth and data,

Emerging Technology

- Blockchain and Cryptocurrency
  - A chain of blocks, each block has 4 compartments with different datasets for each compartment.
  - Decentralized ledger of transactions.
  - Key terms : Hash number, Public Key, Private key.
  - Old Entries can't be deleted, anyone can make new entries. Change resistant and hack proof through hash algorithms.
  - How to establish accountability?
    - Proof of work : mining. Problem : environmental issues, 51% attack.
    - Token based economics and reward for mining.
    - Proof of stake : no competition so no environmental problem. Miners are replaced as validators.
    - Proof of authority : consensus protocol is based on value of identity.
  - Related ideas
    - Presidio principles.
    - Smart Contracts.
    - Quantum Computer may break the crypto algorithm.

Emergent Technology Part 1

- **Emerging Technology**
  - Quantum Computing
    - Superposition
      - Zero and One at the same time.
      - Shrodringer's cat
    - Entanglement
      - Change in one leads to change in the other regardless of the distance between them
      - Superlinkage
      - Example of burgers
    - Qubits rather than normal bits
    - Quantum Supremacy : Sycamore is much faster than any supercomputer.
    - Advantages and Applications : think broad
    - Quantum key distribution : chinese demonstrated quantum computing for cryptography.
      - Future proof cryptography
  - Supercomputers
    - Measured in Floating point operations per second.
    - Work on the principle of parallelism.
    - Fastest Supercomputer : frontier, second fastest : fugaku.
    - National Supercomputing network and national knowledge network,
    - India's Supercomputers : Param Shivay, Param Shakti, Param Brahma, Param Ganga, Pratyush, Mihir, Param siddhi

Emergent Technology Part 2

Emerging Technology part 2

- Digital Signature and Digital Ceritificate
  - Digital Signature : ensures that data entered on blockchain.
  - Use of Hash Algorithm and private key
  - Digital certificate : solves problem of Public key.
  - Digital Certificate is issued by a trusted third party. It not only verifies the identity of the owner but also verifies that owner owns the public key.
- Other Important Technology (just know the meaning)
  - E Sim : Embedded Sim.
  - D2M : broadcast direct to mobile without active internet connection.
  - CrysXPP : using ML to predict the material properties of new materials.
  - Facial Recognition System : confirming the individual's identity using their face. Computer algorithms map unique facial landmarks and convert them to a faceprint.
  - India Stack : Open APIs by India.

Emergent Technology Part 2

Emerging Technology part 2

- **Edge Computing**
  - Data to be analysed locally leading to less latency.
- **4th IR**
  - IoT, Big data, cloud computing, AI, 3D printing, autonomous vehicles.
  - Big data : use of advanced analytics against very large diverse data sets beyond ability of traditional databases.
  - 6V : Velocity, Volume, Value, Variety, Veracity, Variability.
- **Additive Manufacturing**
  - 3D object is built with computer assisted design model by successively adding layers rather than subtractive manufacturing.
  - Power Deposition, Resing curing, Filament Fusing.
  - Terms : .STL file, slicing software.
- **Software as a service**
  - On demand access to application software where user is charged per usage. User need not be concerned about installation / configuration.

Emerging Technology part 2

- **Dark NET**
  - Part of the internet not accessible to traditional search engines like google or traditional browsers like safari.
  - Non Standard Communication protocols which make it inaccessible to ISPs.
  - Need to specific browsers like TOR.
  - Itself part of deep web that is a broader concept.
- **RFID**
  - Tag that can exchange data with an RFID reader through radio waves. Antenna broadcasts energy to tag, which returns a back scatter.
  - One way communication.
  - Low frequency, High Frequency and Ultra High frequency.
  - Does not require a line of sight unlike QR code.
- **NFC**
  - 2 way radio communication at maximum distance of 10cm. RFID is used in NFC.
  - Idea of electromagnetic Coupling.

Emergent Technology Part 2

Emerging Technology part 2

- Satellite Based Internet
  - Usually, internet is wired.
  - Use of Satellites to connect ISPs is satellite based connectivity.
  - Satellite could be placed in LEO or GEO
  - Hybrid model or Direct to Satellite model.
  - Benefits and Challenges as mentioned in class.
- Virtual Private Network
  - Method to give online privacy and anonymity. Provides user access to organisation's network while maintaining the security of information provided.
  - Types : SSL, Site to Site, Client to Server.
- LTE and VoLTE
  - LTE : long term evolution : 4G but voice is routed to older 3G or 2G
  - VoLTE : Voice is converted to data packets and carried over 4G, allowing 4G access even during calls.

Emergent Technology Part 2

Emerging Technology part 2

- **Chat GPT**
  - Generative Pre Trained Transformer developed by OpenAI based on NLP. Trained to predict what comes next and can have a conversation.
  - DALL E - Text to image generator
  - Google : LAMDA chatbot
  - Human AI collaboration : Supportive, Reimaginative, Replacement.
  - Applications of AI : various sectors, think broad.
  - Indian Initiatives and GPAI
- **Metaverse**
  - Blockchain based games and applications with a play to earn model. You get rewarded for being invested in the game.
  - Uses Extended Realities : Virtual Reality, Augmented Reality, Hologram, Avatar.
  - VR : closes the world, AR : enhanced version of physical world, Hologram : 3D objects by interference, Avatar : Representation of individual in virtual world.

IPR and Role of Scientists

IPR and Role of Scientists.

- **IPR**
  - Rights given to persons over the creations of their minds
  - Copyright : Literary and Artistic works, 60 years.
  - Patent : exclusive right of invention, country specific, product or process, 20 years.
    - List of Non Patentable items
    - Compulsory Licensing
    - Parallel Import
    - Evergreening of Patent
    - TKDL
  - Utility model : less strict than patent, between 6 and 15 years.
  - Trademark : distinguishes goods or services of an enterprise from another, initially registered for 10 years.
  - Industrial design : 3d features, 15 years.
  - Geographical Indications : 10 years, for specific geographically attributable products.
  - Trade secrets : Remain Valid as long as independently discovered.

IPR and Role of Scientists

IPR and Role of Scientists.

Nobel prize in Medicine / Physiology

- Swedish Geneticist Svante Paabo
- Sequenced genome of neanderthal and used mitochondrial DNA analysis to understand human evolution and movements,
- Use of Mitochondrial DNA as it does not get influenced by only 1 parent.
- Related Term : denisovans

Standard Model Of Particle Physics

- Finding the fundamental particles of the universe.
- Further division of neutron and proton was possible,
- 6 types of quarks, 6 types of leptons, 4 fundamental forces but gravity is not in the standard model.
- Particles responsible for forces : Gluon, Photon, W/Z boson.
- How do these particles get mass? : explained by higgs boson first detected at the large hadron collider.

IPR and Role of Scientists

- **IPR and Role of Scientists.**
  - **Nobel Prize in Chemistry**
    - Click chemistry and Bioorthogonal reactions.
    - Gets past the problem of unwanted products coming up due to artificial copying of natural processes.
    - Important : Copper Catalysed azide - alkyne cycloaddition.
    - Bioorthogonal chemistry : using this process in living beings without copper ions. This helps us tag glycans.
    - Applications : develop pharmaceuticals, herbicides, understand complex biological process, drug development and targeting.
  - **Nobel Prize in Physics**
    - Quantum Entanglement, Quantum Teleportation, Quantum Entanglement swapping and theoretical work on bell inequalities.
    - Related term : no cloning no deleting theorem.
    - Entanglement Swapping